Arabic Sign Language Translator

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Introduction

Communication barriers between deaf and hearing individuals remain a significant challenge in Arabic-speaking communities. Arabic Sign Language (ArSL) lacks affordable, real-time digital translation tools — especially ones that work in Egyptian and regional dialects.
This project presents an AI-powered system that recognises Arabic Sign Language letters from live webcam video and translates them into Arabic text and speech in real time.

32 ArSL Letters	**28.4K** Training Images	**96.2%** Test Accuracy

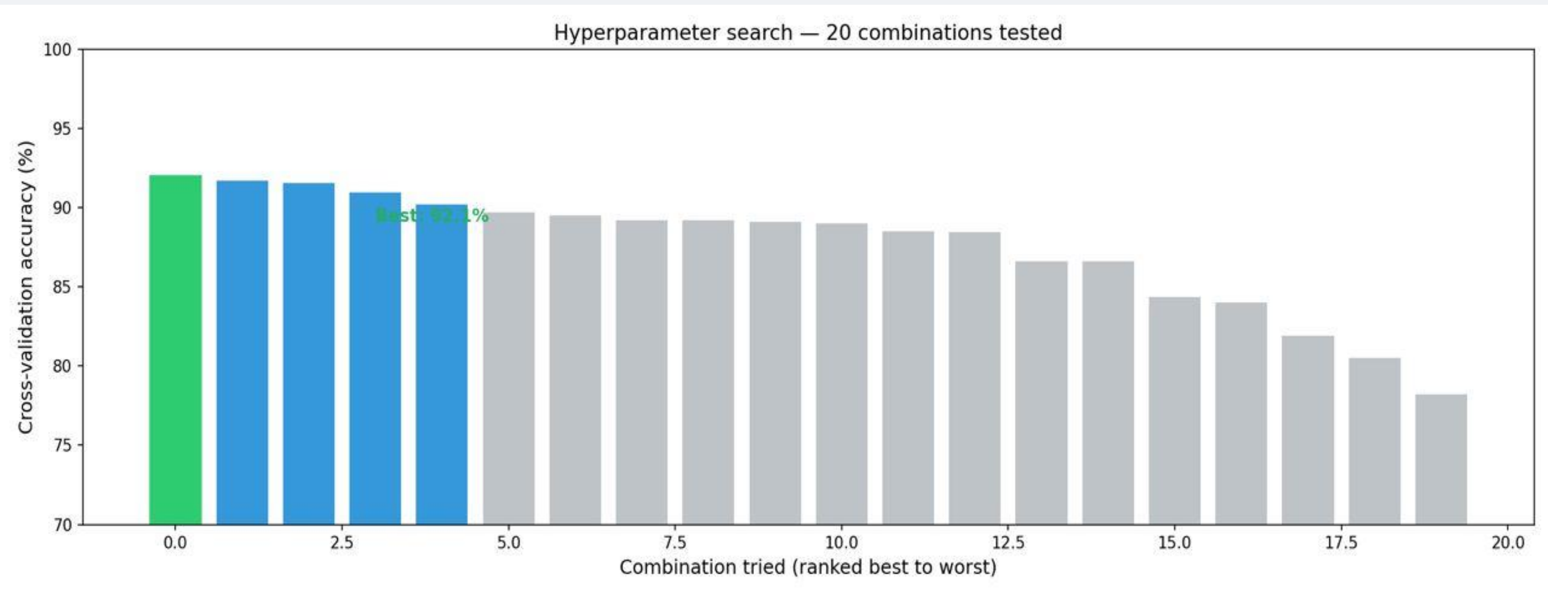
Problem Statement & Objectives

PROBLEM

- No real-time ArSL tool exists
- 6.7× class imbalance (gaaf: 46 vs sheen: 308)
- 32 visually similar letter classes
- Existing AI tools lack Arabic Sign Language support

OBJECTIVES

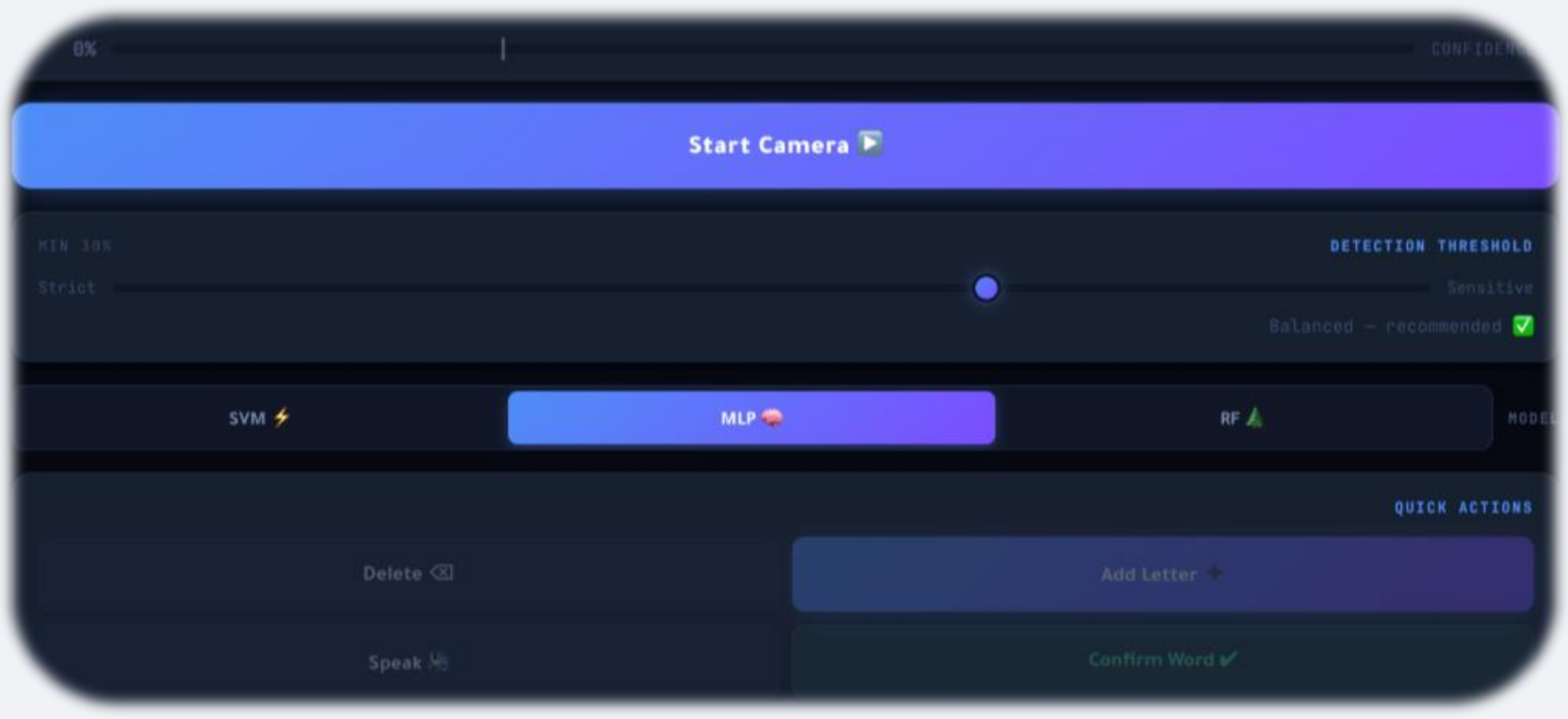
- Detect all 32 ArSL letters in real time
- Achieve >80% test accuracy
- Build Arabic words letter by letter
- Arabic Text-to-Speech output via gTTS
- Deploy as a live full-stack web application



RandomizedSearchCV tested 20 parameter combinations.
Best validation accuracy achieved: 92.1%.

Proposed Work

- •Developed a real-time Arabic Sign Language translation system using AI and computer vision.
- •Used MediaPipe Hands to extract 21 hand landmarks and generate 63 spatial features.
- •Applied preprocessing and wrist-based normalization for position-invariant detection.
- •Solved dataset imbalance using SMOTE augmentation techniques.
- •Trained and compared multiple machine learning models including Random Forest, SVM, and MLP.
- •Achieved the highest performance using the MLP model with 81.24% accuracy.
- •Built a full-stack web application using React + Vite frontend and FastAPI backend.
- •Integrated real-time webcam prediction, Arabic text generation, and speech output.



Real-time prediction interface with adjustable confidence threshold and dynamic model selection (SVM, MLP, RF).

Conclusion

- •The proposed system successfully translates Arabic Sign Language letters in real time.
- • MediaPipe landmarks provided fast and lightweight feature extraction compared to CNN-based approaches.
- •SMOTE balancing improved model performance and reduced class imbalance issues.
- •The MLP model achieved the best overall accuracy and stability.
- •The web-based deployment enabled accessible real-time communication support for deaf and hearing users.